

A PROJECT for GENERAL CHEMISTRY STUDENTS: COLOR TONING of PHOTOGRAPHIC PRINTS

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A GROUP of freshmen, wishing to take part in the Intercollegiate Science Conference,* undertook the color toning of some photographic prints, a project suggested by an article appearing in *Nature Magazine*.¹ Most of the work was done by four students,† one of whom wrote and presented a paper describing the methods used and explaining the chemical reactions involved. The work consisted in preparing solutions, toning and drying the prints, and mounting them on posters, and took about four evenings' time. Since the effects produced were pleasing and the chemical reactions involved were among those usually studied in a general chemistry course, or similar in character, it was thought that others teaching or studying chemistry might be interested in an account of the work.

As is generally known, the process of developing and fixing a print leaves a silver image on the paper. The toning processes used consist of converting the silver of the image into silver sulfide (sepia tones), cupric ferrocyanide (red or reddish-brown tones), ferric ferrocyanide (Prussian blue), or a mixture (green)² of iron (blue) and vanadium (yellow) ferrocyanides. The color can be suited to the particular picture at hand, or an interesting series of prints can be obtained by taking several copies of the same picture and toning all but one to the various colors desired.

A half-dozen prints of each of four campus scenes were made, in postcard size, by a commercial photographer. In the case of each photograph, one print was kept in its original form and the others subjected to the various toning processes discussed in this paper. After toning, each set of prints was mounted on a large sheet of cardboard. The four posters thus obtained showed in a striking way the different effects possible. A fifth poster consisted of a sheet of cardboard with labeled vials attached, each vial containing the precipitate corresponding to one of the tones on the prints.

* A one-day program of papers and exhibits by students in eleven New England colleges, held annually since 1931.

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¹ BROWN, SAMUEL, "Photographic notes," *Nature Mag.*, **25**, 50 (Feb., 1935). Mr. Brown was most helpful in giving advice and also in furnishing literature on photography.

² NEBLETTE, C. B., "Photography, its principles and practice," D. Van Nostrand Co., New York City, 1927, ch. XX.

The toning baths were prepared according to formulas given in a pamphlet issued by the Gevaert Company.^{3,4} In each case the print was soaked for a short time in water, then immersed in the toning bath, with constant rocking, and finally washed in running water and dried. The work was carried on under artificial light or diffuse daylight.

From the three methods for obtaining sepia prints described in the Gevaert pamphlet, the sulfide re-developer or "indirect sulfur" process was chosen because it could be used at room temperature. In this method the prints are first "bleached" in a bath containing potassium bromide and potassium ferricyanide. The latter compound is reduced to potassium ferrocyanide, the silver being converted to silver ferrocyanide, then to yellowish silver bromide. After a short washing to remove the potassium ferrocyanide, the print is immersed in a bath of dilute sodium sulfide, which changes the silver bromide to the less soluble silver sulfide.^{5,6} The final color and quality of the print depend on the condition of the baths and the length of immersion. The color varies from light brown, when fine particles are formed, to black tones, when the silver sulfide is formed in coarser grains. If the print is removed from the first bath before bleaching is complete, a darker sepia tone results. Moreover, the technic used in the exposure and development of the print is of great importance. The most satisfactory results are obtained when prints are fully exposed and developed before toning; over-exposed and under-developed prints yield "washed-out" tones.⁷ A sulfide print is quite permanent, more so, if anything, than the original silver image.⁸

³ "Formulas for Gevaert films, plates, papers," The Gevaert Company of America, Inc., 423-439 West 55th Street, New York City, 1933, pp. 23-6.

⁴ Slightly different toning formulas are given in the following: FRAPRIE, FRANK R., "How to make prints in colors," American Photographic Publishing Co., 428 Newbury St., Boston, 1916, pp. 27-32.

⁵ MEES, C. E. K., "Fundamentals of photography," 5th ed., Eastman Kodak Co., Rochester, N. Y. (Mar., 1931), pp. 98-9.

⁶ FLINT, WILLIAM RUTHVEN, "Chemistry for photographers," 2nd ed., American Photographic Publishing Co., Boston, 1916, p. 122.

⁷ BROWN, SAMUEL, private communications to the author.

⁸ SEYEWETZ, A., "Les methodes actuelles de virages en diverses couleurs des papiers photographiques aux sels d'argent," *Chimie et Industrie*, **5**, 3-10 (1921).

The other tones were obtained by the use of baths which contained potassium ferricyanide as oxidizing agent, a metallic salt whose ferrocyanide is insoluble, and an organic acid (or its salt) which, by complex ion formation, retards the deposition of the colored ferrocyanide.⁷ In each case the bath was prepared in three parts which were mixed just before use. As in sulfide toning, the preliminary reaction consists in oxidation of the silver to silver ferrocyanide with accompanying formation of potassium ferrocyanide. The potassium ferrocyanide then gives a colored ferrocyanide with the metallic salt.

The blue toner consists of: part "A", a solution of ferric ammonium citrate; part "B", which contains potassium ferricyanide; and part "C", which is diluted acetic acid. The print is rocked in the bath until the image is bright blue, then washed in clear water until the highlights are clean. Rather light prints and short toning give the best results, since the blue toner slightly intensifies the image. Prolonged toning is likely to tint the highlights; this can be remedied by washing in a very weak alkaline bath.⁷

The choice of ferric salt is important, ferric chloride, for example, not serving since it forms silver chloride. It should be completely free of ferrous salt or the bath will contain ferrous ferricyanide, which will tint the whites of the image.⁸

The red toner consists of the following: part "A", potassium citrate solution; part "B", copper sulfate solution; and part "C", potassium ferricyanide solution. Potassium citrate prevents the precipitation of cupric ferricyanide without causing, as would sodium citrate, solution of cupric ferrocyanide.⁸ Short immersion gives a warm black, while prolonged immersion continues the reaction, producing a fine red chalk shade. Various intermediate shades of purple, brown, and red may be obtained. Thorough washing follows. "Vigorous" prints are needed for attractive tones.

The green toner is a mixture prepared from: part "A", a solution of oxalic acid, ferric chloride, and ferric oxalate; part "B", potassium ferricyanide solution; part "C", a solution of vanadium chloride and hydrochloric acid. The print is toned until deep blue, then washed until the tone changes to green. Short toning and washing are best; longer toning gives a brighter green, longer washing gives a yellower tone. A very strong print is needed for good results. On account of partial washing away of the image, these were the least satisfactory prints obtained. However, the prints used in the toning were rather weak at the start. An "improved vanadium toning bath" which has not been tested in our laboratory is described by van Beugen.⁹ A yellowish stain in the whites may be removed by immersion in ammonium thiocyanate solution.

The stability of iron or vanadium toned prints over a period of years is doubtful. However, care in washing

all soluble products out of the prints⁶ lessens the tendency toward deterioration. A thin coating of varnish or gum¹⁰ or of a one per cent. benzene solution of paraffin¹¹ has been suggested as protection against atmospheric action.

It is possible to tone prints with more than one color. This may be accomplished by painting the parts not to be toned with a spirit varnish, such as white shellac. After toning in the first bath the varnish is removed from the untuned part of wood alcohol. The toned portion is next protected by varnish and the toning process continued in the second color.¹ It is well to varnish the back of the print, to prevent penetration of the toning solution from behind.

Numerous other toning processes are described in the literature, involving, for example, the use of uranium, molybdenum, cobalt, lead,^{8,10} and selenium¹² compounds as well as the gold and platinum processes commonly used in commercial photography. The references which have been cited in this paper contain a great deal of information which anyone undertaking the toning of prints would find of value. Various precautions generally mentioned in the literature include the following: in preparing the original print, avoid the use of an old fixing bath since this may lead to stains in toning, resulting from the silver compounds present in a used bath; completely remove all hypo by washing, to prevent its action on the silver ferrocyanide formed during the toning process; use matt or semimatt paper in preference to glossy papers, which are more difficult to work with and give less uniform results; in all toning processes, use "vigorous" prints (that is, prints with good contrast). For a good blue-toned picture, the print should be made somewhat lighter than a print which is to be toned any other color. It is also well to experiment on a few old prints to learn proper timing for the color tone desired.

It must be recognized that because of the complexity of photographic reactions in general and the great number of variable factors, the chemical reactions taking place on a photographic print are probably not in all cases exactly the same as the corresponding test-tube reactions. This explains a divergence of opinion about the products formed among some who have investigated ferrocyanide toning processes.¹³

¹⁰ WALL, E. J. AND F. J. MORTIMER, "Dictionary of photography," 11th ed., Iliffe and Sons, Ltd., London, 1927, p. 639.

¹¹ PIZZIGHELLI, G., "Leitfaden der praktischen Photographie," 14th ed., by Paul Hanneke, Wilhelm Knapp, Halle-Saale, 1919, p. 277.

¹² VAN BEUGEN, E., "Tests with selenium toning bath," *Focus*, 20, 72-4 (1933) (through *Chem. Abstr.*, 27, 2642¹ (1933)). Also later papers by the same author.

¹³ See, for example, (in addition to the references already given) the following: (a) NAMIAS, R., "Toning with ferrocyanides," *Atelier Phot.*, 15, 4 (Dec., 1907) (through *Chem. Abstr.*, 2, 1390⁵ (1908)); (b) LUMIERE, A. AND L., AND A. SEYEWETZ, "Composition of silver images toned with metallic salts," *Wiener Mitt.*, 13, 254 (June) (through *Chem. Abstr.*, 2, 2655⁷ (1908)); (c) LEHMANN, E., "Copper toning as a means of mordanting for dye images," *Phot. Ind.*, 1924, 526-7 (through *Chem. Abstr.*, 19, 218¹ (1925)).

⁹ VAN BEUGEN, E., "Improved vanadium toning bath," *Focus*, 20, 39 (1933) (through *Chem. Abstr.*, 27, 2642¹ (1933)).